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CUSTOMER SEGMENTATION & BEHAVIOR ANALYSIS

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PORBLEM STATEMENT

Businesses struggle to segment their customers effectively, leading to inefficient marketing strategies and poor personalization. This project aims to analyze customer purchasing behavior and segment them into distinct groups using machine learning techniques. By clustering customers based on their income, spending patterns, and purchase frequency, businesses can enhance targeted promotions, retention strategies, and product recommendations.

Dataset Attributes

A Customer_ID → Unique identifier for each customer

B Age → Age of the customer

C Gender → Male, Female, Other

D Income → Annual income in USD

E Spending Score → Index (1-100) indicating spending habits

F Purchase Frequency → Number of purchases per month

G Product Category → Preferred shopping category

H Total Spending → Amount spent in the last 6 months

CUSTOMER SEGMENTATION BASED ON SPENDING & INCOME

In [19]:

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x=df['Income'], y=df['Spending'], hue=df['Cluster'], palette="deep")
plt.title("Customer Segmentation based on Spending & Income")
plt.xlabel("Annual Income")
plt.ylabel("Total Spending")
plt.show()
```

Goals: Segment customers based on income and spending patterns to optimize marketing and retention strategies.

Insights: High-income individuals don't always have high spending; segmentation enhances personalized engagement.

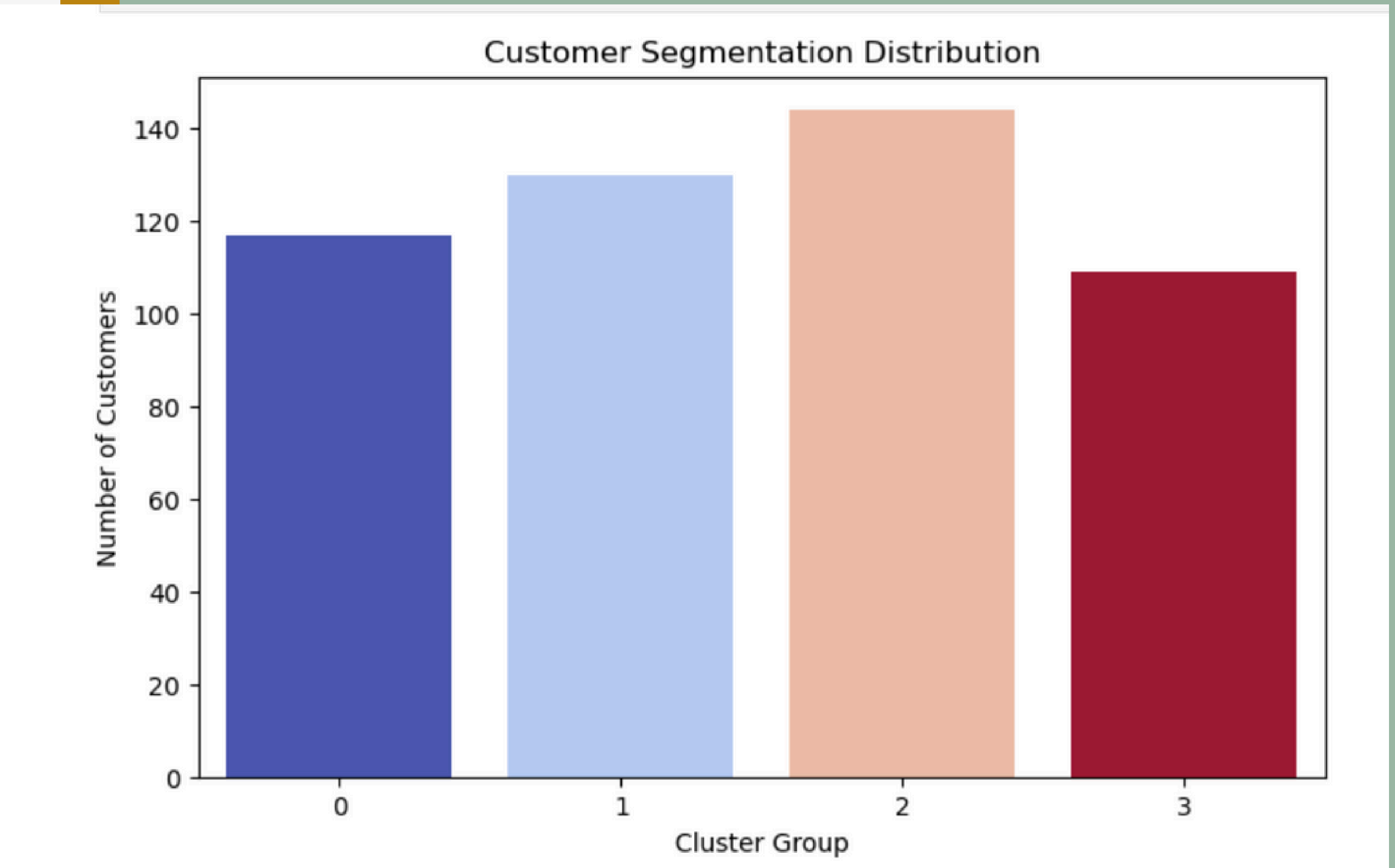
CUSTOMER SEGMENTATION DISTRIBUTION

In [37]:

```python
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))
sns.countplot(x=df['Cluster'], hue=df['Cluster'], palette='coolwarm', legend=False)
plt.title("Customer Segmentation Distribution")
plt.xlabel("Cluster Group")
plt.ylabel("Number of Customers")
plt.show()
```

Goal: Understand how customers are grouped within clusters.

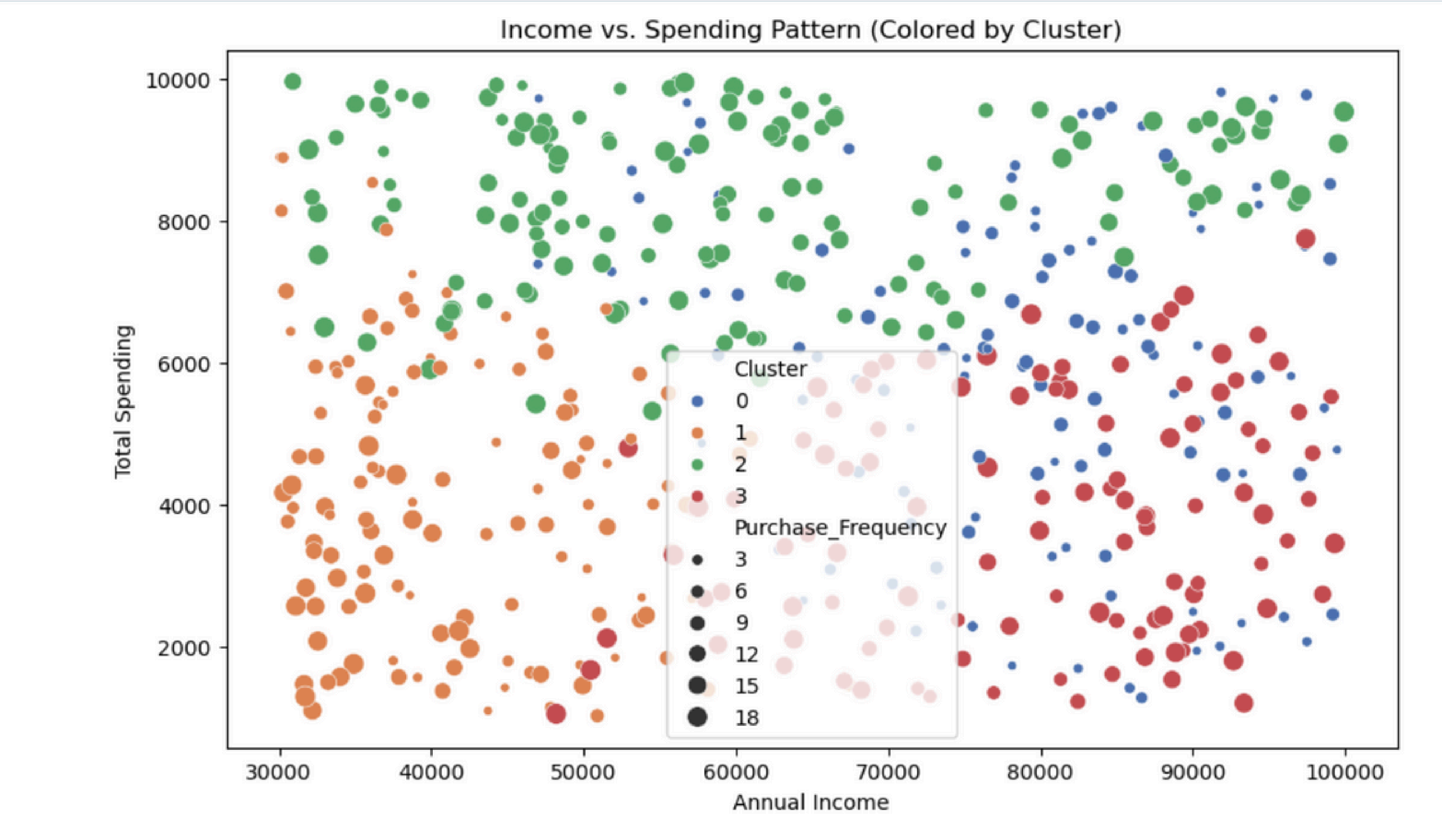


Insight: See which segment has the highest number of customers, helping in marketing decisions.

SPENDING PATTERN VS. INCOME

Goal: Identify how spending habits vary across different income levels.

```
In [27]:
plt.figure(figsize=(10, 6))
sns.scatterplot(x=df['Income'], y=df['Spending'], hue=df['Cluster'], palette="deep", size=df['Purchase_Fr
plt.title("Income vs. Spending Pattern (Colored by Cluster)")
plt.xlabel("Annual Income")
plt.ylabel("Total Spending")
plt.show()
```

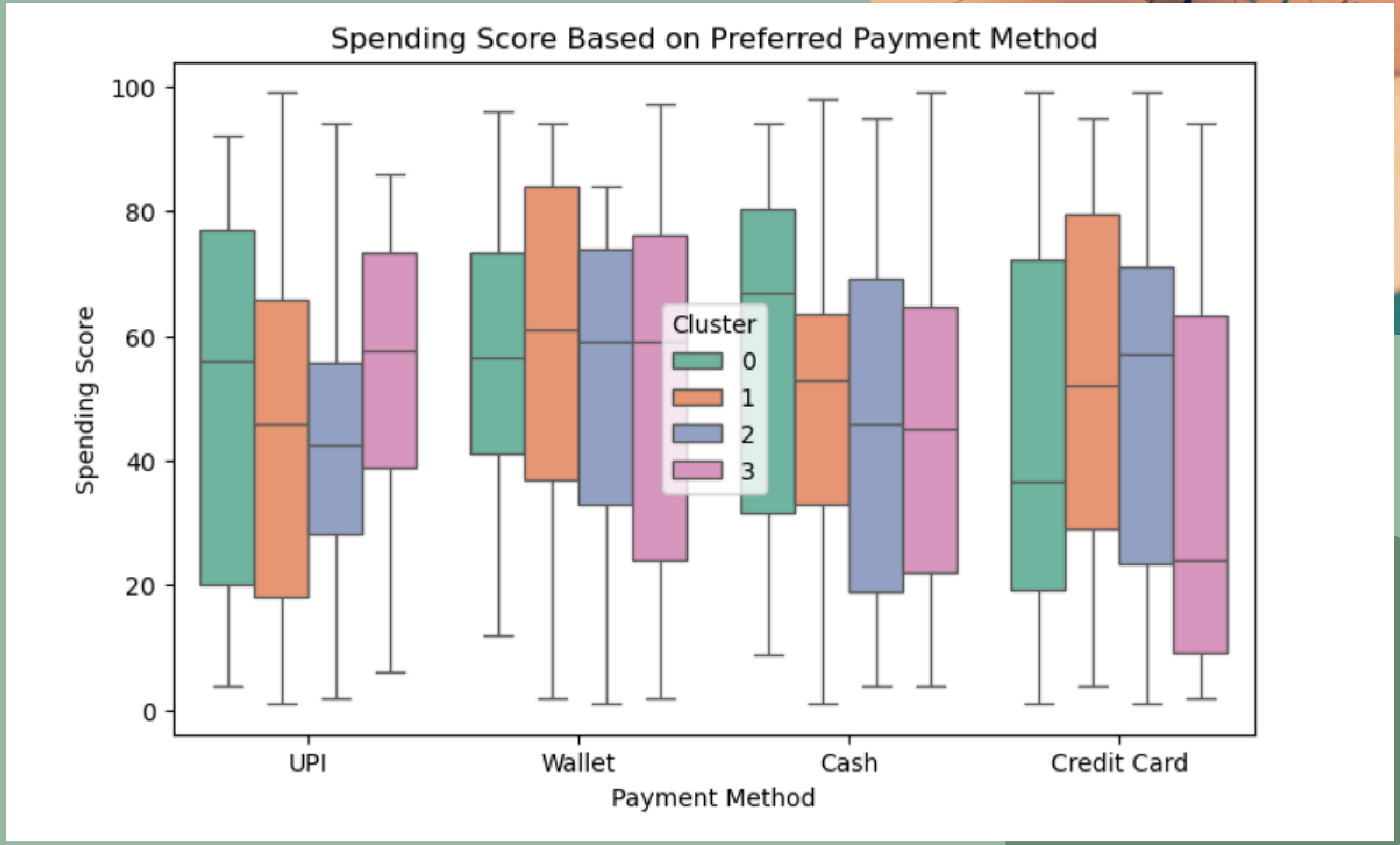


Insight: Higher-income groups may not always be high spenders. Helps in targeted promotions.

PREFERRED PAYMENT METHODS ACROSS CLUSTERS

In [35]:

```
plt.figure(figsize=(10, 6))
sns.boxplot(x=df['Cluster'], y=df['Purchase_Frequency'], hue=df['Cluster'], palette="coolwarm", legend=Fa
plt.title("Purchase Frequency Distribution Across Customer Segments")
plt.xlabel("Customer Cluster")
plt.ylabel("Purchase Frequency")
plt.show()
```

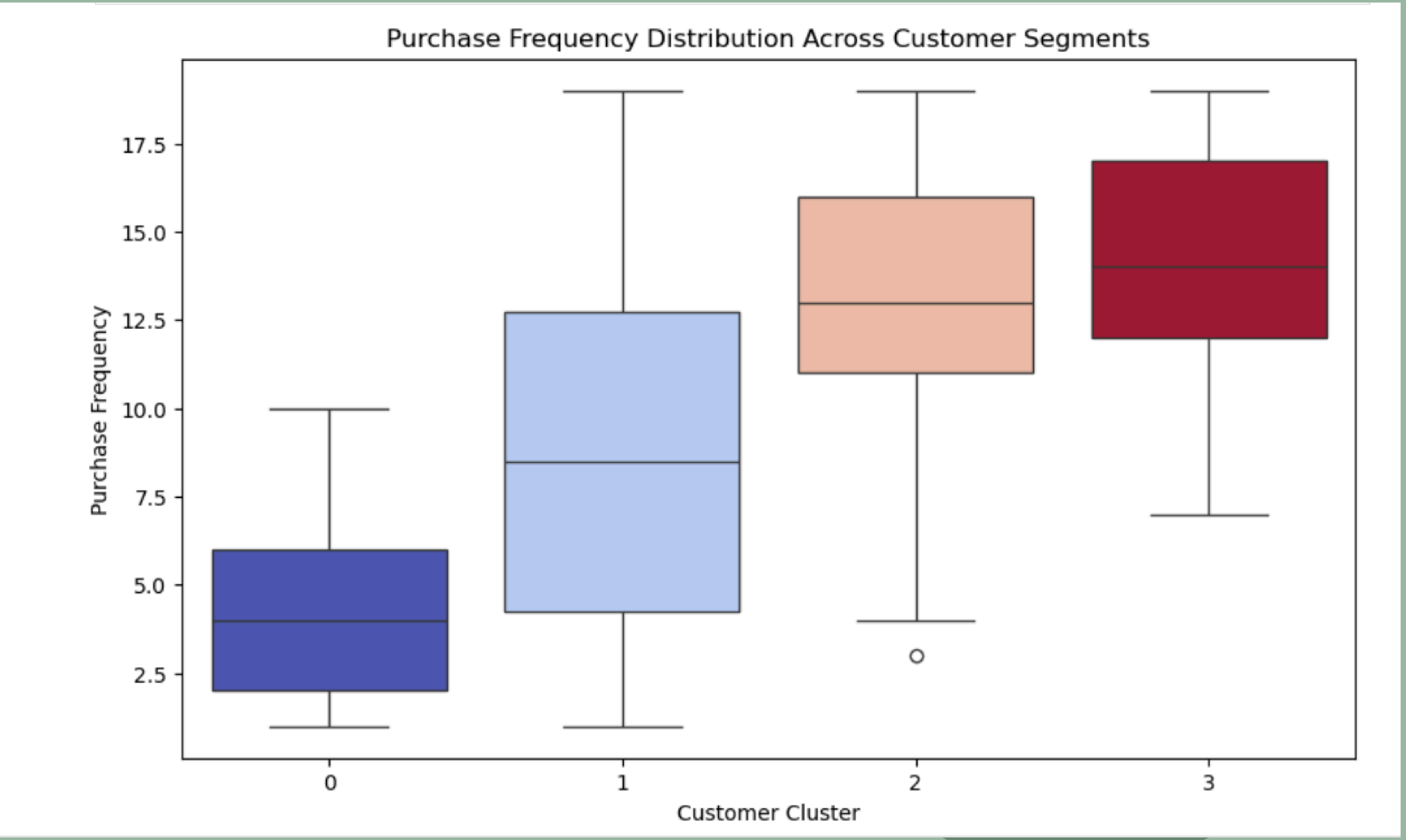


Goal: Understand how different customer segments prefer to pay.

Insight: Helps businesses prioritize payment options for different customer groups.

CUSTOMER ENGAGEMENT ANALYSIS

```
In [35]:
plt.figure(figsize=(10, 6))
sns.boxplot(x=df['Cluster'], y=df['Purchase_Frequency'], hue=df['Cluster'], palette="coolwarm", legend=Fa
plt.title("Purchase Frequency Distribution Across Customer Segments")
plt.xlabel("Customer Cluster")
plt.ylabel("Purchase Frequency")
plt.show()
```



Goal: Compare purchase frequency trends in different clusters.

Insight: Identify loyal customers who frequently shop and need exclusive loyalty programs.

BUSINESS IMPACT → HOW SEGMENTATION IMPROVES DECISION-MAKING

High-spending customers can be targeted for premium offers.

Occasional buyers might need personalized promotions to increase engagement.

Budget-conscious consumers can benefit from discounts and loyalty programs.

INSIGHTS →

- Spending Behavior Patterns
- Customer Payment Preferences
- High-Value vs. Low-Engagement Customers
- Potential Business Actionables

CONCLUSION & FUTURE SCOPE → KEY TAKEAWAYS AND NEXT STEPS

THREE MAIN TAKEAWAYS:

Segmentation helps businesses improve personalized marketing.

High-value customers can be identified for loyalty programs.

Future improvements include refining clustering models and real-time tracking.

- How can businesses apply segmentation strategies in real-time?
- How does customer behavior impact product pricing?
- Can additional features improve segmentation accuracy?

THANK YOU

-By Bhavya Balyan